



Discrete viscous dampers for multi-mode vortex-induced vibration control of long-span suspension bridges

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ABSTRACT

Multi-mode vertical vortex-induced vibrations have been observed on several long-span suspension bridges, where several vertical modes of stiffening girder are excited in turn with increasing of wind velocity. This paper proposed a discrete viscous dampers solution and its optimization targeting multi-mode vortex-induced vibration control for long-span suspension bridges. The minimal modal damping ratio was used to evaluate the performance of the damper system. The wind tunnel test verified that the modal damping ratio is an alternative to the resonance response amplitude and can be obtained by complex modal analysis. The ternary search algorithm was adopted to find the optimal damper parameter for the Yingwuzhou Yangtze River Bridge, which is a suspension bridge that suffered excessive vortex-induced vibrations. Using linear damper elements to connect the bridge tower and the bridge girder directly turned out to be a feasible method of increasing multiple modal damping ratios. Four torsional eddy current dampers have been installed in the Yingwuzhou Yangtze River Bridge, based on the numerical optimization results calculated in this paper. Although the measured modal damping ratios were smaller than the numerical results, the structural damping increased by three times after dampers were installed, and the requirement of the current specification was satisfied.

1. Introduction

Long-span bridges are susceptible to wind-induced vibration, such as flutter, buffeting, and vortex-induced vibration (VIV), due to their low frequencies of oscillation and low mechanical damping (Simiu and Scanlan, 1996). The VIV occurs when the vortex shedding frequency coincides with a natural structural frequency (Ehsan and Scanlan, 1990). For long-span suspension bridges, there exist several closely spaced vertical modes, and these modes, in order of natural frequency, may be excited by VIV in turn with an increase of wind velocity. This is the multi-mode vertical VIV problem, which usually occurs at a low wind speed that is commonly encountered with a high-frequency mode (Diana et al., 2006). Multi-mode VIV events have been reported in many suspension bridges in actual use worldwide. For example, the Xihoumen Bridge experienced vertical VIV events with frequencies of 0.183 Hz, 0.276 Hz, 0.327 Hz, 0.378 Hz, 0.435 Hz, and 0.495 Hz (Li et al., 2017). The Great Belt Bridge experienced vertical VIV events with frequencies of 0.130 Hz, 0.209 Hz, and 0.242 Hz (Frandsen, 2001). The Humen

Bridge experienced vertical VIV events with frequencies of 0.13 Hz, 0.23 Hz, 0.27 Hz, and 0.37 Hz (Zhao et al., 2022). Though the VIV is self-limiting and the response amplitude is usually moderate, it can affect serviceability or cause fatigue problems at a low wind speed even below 10 m/s. Therefore, VIV mitigation is necessary in practice and has drawn extensive attention in the field of bridge wind engineering during the last decades.

Countermeasures to suppress VIV can be divided into two types: aerodynamical and mechanical. Theoretically, aerodynamical countermeasures can fundamentally eliminate the VIV by interfering the flow field around the main girder to prevent the formation and shedding of the periodic vortex. To achieve this aim, configurations of some non-structural components (e.g., crash barriers (Kubo et al., 2002) and maintenance rails (Laima et al., 2018)) can be adjusted. Alternatively, installing additional components such as winglets, guide vanes, and deflectors has also been proven to suppress VIV (Larsen et al., 2000; Bai et al., 2020). Due to the intrinsic complexity of fluid-structure interaction existing in the VIV, the performance of aerodynamical

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countermeasures is highly sensitive to the configuration of the main girder (Yang et al., 2016). As a result, for each long-span suspension bridge, the effective aerodynamic countermeasure is generally obtained from sectional model wind tunnel tests through a tedious trial-and-error procedure. Besides, simplifications and model scaling in wind tunnel experiments have been reported to have unexpected influences on aerodynamic characteristics (Schewe and Larsen, 1998; Jones et al., 1995; Larose and D'Auteuil, 2006). Further examination is required to check whether the aerodynamical countermeasures can maintain the same effect in the actual structure. Moreover, for bridges in service, the construction process of applying aerodynamical measures is complex and expensive, which may also interrupt normal operation and cause widespread social concern.

Since the VIV of bridges is highly sensitive to the structural damping level, mechanical countermeasures are also frequently used to suppress VIV of long-span bridges by installing certain energy dissipation devices to supply additional modal damping ratios. For long-span continuous steel box-girder bridges, tuned mass damper (TMD) is the most popular and effective mechanical countermeasure to control the excessive vibration due to VIV, and it has been employed in many bridges worldwide, such as the Trans-Tokyo Bay Crossing Bridge in Japan (Fujino and Yoshida, 2002), the Rio-Niterói Bridge in Brazil (Battista and Pfeil, 2000), the Chongqi Bridge in China (Sun et al., 2020) and the Volgograd Bridge in Russia (Weber and Mašlanka, 2012). A traditional TMD works by transferring the vibration energy of the host structure to an additional mass and dissipating it, which has to be finely tuned to a particular control mode and typically requires an additional mass between 0.5% and 2.0% of the controlled modal mass in applications. It is relatively easy to achieve this goal for TMD utilized on a continuous steel box-girder bridge, where the VIV event is often dominated by only one or two modes. However, there exist some inherent limitations for long-span suspension bridges. Firstly, the control bandwidth of TMD is narrow, and multiple TMDs should be used to control multi-mode vertical VIV. The additional mass of TMD required to control VIV in a single mode generally weighs hundreds of tons for long-span suspension bridges. As a result, a total mass of TMDs weighing thousands of tons is needed to solve the multi-mode vertical VIV problem of long-span suspension bridges. Secondly, due to the low frequency of long-span suspension bridges, the tuned TMD stiffness is extremely low, resulting in a large static deformation of the springs, which is difficult to manufacture and install on existing bridges. The static deformation of TMD springs can be estimated by $\Delta \approx 0.25/f_n^2$ (f_n is the modal frequency of the bridge to be controlled) (Huang et al., 2019). For an example, the static deformation corresponding to the frequency 0.2 Hz is 6.25 m, which could be too large.

To reduce the large static deformation of the spring of the classical hanging-type TMD, one way is to use the lever-type TMD (Gu et al., 2002). The other is to employ the TMDI, in which an inerter is added to a traditional TMD between the secondary mass and the structure (Giaralis and Petrini, 2017). The static stretching of the spring of TMDI is much smaller than that of the traditional TMD, due to the dynamic negative stiffness contributed by the inerter (Xu et al., 2020; Weber and Mašlanka, 2012). However, compared with the TMD, the control efficiency of TMDI could be reduced (Xu et al., 2019) unless the inerter is installed at a position far from the mass stock (Dai et al., 2021), which may face practical challenges. Moreover, in consideration of the nonlinearity of the inerter, the effectiveness of the TMDI would be further reduced (Xu et al., 2022).

Apart from the dynamic absorbers such as TMD and TMDI, the vibration of civil engineering structures such as cables and beams can also be controlled by discrete dampers directly. Krenk (2004) established a state space framework for calculating the dynamic characteristics of continuous structures with discrete dampers. Then, Main and Krenk (2005) developed an approximate solution to estimate the complex modal of a discrete system with discrete dampers and examined the

damper effect on a 10-story frame model. Yang et al. (2021) applied two viscous dampers to control the cable vibration and verified the multi-mode control performance of the discrete dampers. Möller et al. (2019) proposed a damping system for flutter control of a long-span suspension bridge, which is composed of a spring and a viscous damper in parallel, connecting the tower with bridge main cables via a pre-tensioned cable. Recently, Zoltowski et al. (2022) applied an extended viscous damper to control pedestrian vibrations. However, to the authors' best knowledge, the study on the performance of discrete dampers for multi-mode VIV control of suspension bridges has not yet begun. The question of how to optimize damper parameters to control multi-mode vibrations remains unclear.

Since VIV control of long-span suspension bridges is still a challenging issue. This paper hence proposed a discrete viscous dampers solution for multi-mode VIV control of long-span bridges. The solution is suitable for VIV control of almost all suspension bridges without the additional mass. With the help of the complex modal analysis, the optimal parameters can be easily obtained. The construction of the discrete damper system for a long-span suspension bridge was discussed in Section 2. The governing equation of the complex modal analysis and a multi-mode optimization process of structure with discrete dampers was proposed in Section 3. In Section 4, the controlled performance of the discrete damper system was verified by the application in the VIV control of the Yingwuzhou Yangtze River Bridge. Finally, the conclusions and discussions were briefly addressed in Section 5.

2. Installation scheme of the damper system

Similar to the conception of the vibration control of stay cable using a damper attached near the anchorage, a feasible option for a long-span suspension bridge is connecting the bridge girder and the bridge tower. For the floating suspension bridge, there is a relative modal displacement between the girder and the tower. Damping effects can be achieved by installing the damper directly at the intersection of the girder and tower, as shown in Fig. 1. When there is no relative displacement at the intersection, such as in simply-supported bridges, the position of the damper needs to be far away from the support to ensure the relative displacement, as shown in Fig. 2. A rigid corbel is required when the damper is located beyond the bridge tower. For in-serve bridges, dampers are suggested to be placed under the local stiffeners, such as the diaphragms and the suspender's junctions. Otherwise, additional local stiffeners need to be installed at the damper's junction.

Though fluid viscous damper has been widely used to reduce the dynamic response of large-scale structures (Möller et al., 2019; Chen et al., 2020; Yang et al., 2021; Zoltowski et al., 2022), special considerations should be taken when a viscous damper is employed to supply vertical damping force for a long-span suspension bridge. Besides controlling the vertical VIV, the viscous damper needs to work under different loading conditions, such as the traffic load, the temperature fluctuation, and the seismic loading. Under the action of traffic load or temperature fluctuation, the suspension bridge not only has vertical deflection but also longitudinal horizontal motion, the amplitude of which is usually much greater than that of vertical VIV at the damper position. Under strong earthquakes, the longitudinal motion of long-span suspension bridges may be even larger than the former ones, so a viscous damper with a length of up to several meters may be needed to meet the stroke requirement, which may be difficult to manufacture, hard to maintain, and expensive. In other words, it is preferred to reduce the influence of the longitudinal movement of the girder on the working of the damper. The damper scheme and the stroke capability employed in an actual bridge are shown in Fig. 3. A torsional eddy current damper is developed, which utilizes the crank-connecting rod mechanism to convert the vertical vibration of the bridge into a rotational motion of the torque damper. With the help of the crank-connecting rod mechanism, the longitudinal motion of the long-span suspension bridges can be well met without much increase in the stroke of the damper.

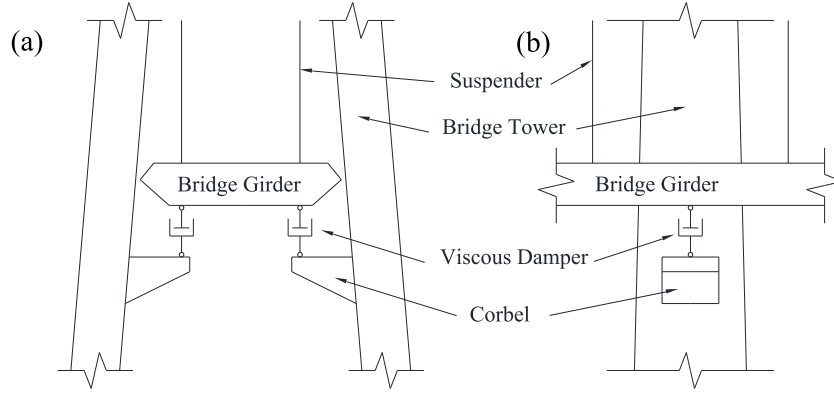


Fig. 1. Schematic of the damper arrangement of the floating suspension bridge at (a) transverse direction and (b) longitudinal direction of the bridge.

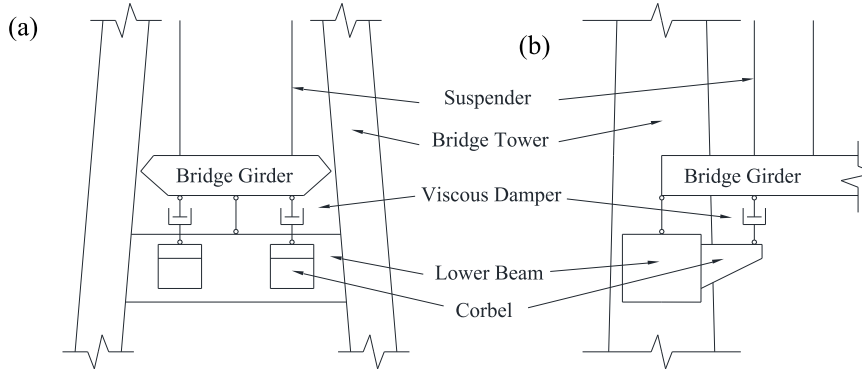


Fig. 2. Schematic of the damper arrangement of the simply-supported suspension bridge at (a) transverse direction and (b) longitudinal direction of the bridge.

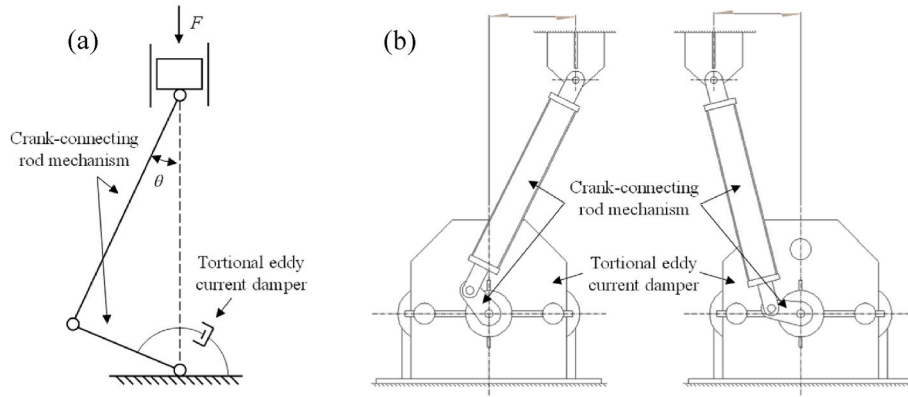


Fig. 3. Illustrations of (a) schematic and (b) stroke capability of the damper mechanism.

Additionally, a torsional eddy current damper does not require sealing and has a longer service life than that of the fluid viscous damper.

3. Governing equations and optimization

3.1. Complex modal analysis and damper's contribution

Based on the complex modal theory, the n -DOF linear motion equation with linear viscous damping can be described in the state-space form:

$$\begin{bmatrix} \mathbf{C} & \mathbf{M} \\ \mathbf{M} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{X}} \\ \ddot{\mathbf{X}} \end{Bmatrix} + \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0} & -\mathbf{M} \end{bmatrix} \begin{Bmatrix} \mathbf{X} \\ \dot{\mathbf{X}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{F} \\ \mathbf{0} \end{Bmatrix} \quad (1)$$

where, submatrices $\mathbf{K}$, $\mathbf{C}$, and $\mathbf{M}$ respectively represent the stiffness, damping, and mass of the linear system; $\{\mathbf{F}\}$ is the external force; vector $\{\mathbf{X}, \dot{\mathbf{X}}\}^T$ is known as the state vector; the dot marks the differentiation of time.

The complex modes are defined by the solution of the homogenous linear equation, that is $\{\mathbf{F}\} = 0$, which is independent of the

characteristic of the load and can be expressed as:

$$\dot{X} = \Psi e^{\lambda t}, \dot{X} = \Psi \lambda e^{\lambda t} \quad (2)$$

where, λ is the eigenvalue of Equation 1, and $\{\Psi, \lambda \Psi\}^T$ is the corresponding eigenvector. It has n pairs of conjugate eigenvalues and eigenvectors. The r -th eigenvalue λ_r and its conjugate λ_r^* are usually rewritten by two real parameters, the modal damping ratio ζ_r and the modal frequency ω_r , that is:

$$\lambda_r, \lambda_r^* = -\zeta_r \omega_r \pm \sqrt{1 - \zeta_r^2} \omega_r i \quad (3)$$

where, $i = \sqrt{-1}$; $\zeta_r = -\text{Re}\{\lambda_r\}/|\lambda_r|$; $\omega_r = |\lambda_r|$. ω_r is an artificially defined frequency that is equal to the undamped frequency ω_{0r} only if the damping matrix C meets the following Caughey-O'Kelly condition (Caughey and O'Kelly, 1965):

$$CM^{-1}K = KM^{-1}C \quad (4)$$

For discrete viscous dampers system, the damping matrix is defined as:

$$C = \sum_j c_j \mathbf{w}_j \mathbf{w}_j^T \quad (5)$$

where, c_j is the viscous damping parameter of the j -th damper, and vector $\mathbf{w}_j$ represents the attachment position and orientation. For example, the vector $\mathbf{w}_1$ corresponds to a damper connecting 3rd and 4th DOF of a 5-DOF system is defined by:

$$\mathbf{w}_1 = [0 \ 0 \ -1 \ 1 \ 0]^T \quad (6)$$

It is obvious that the damping matrix does not strictly match the Caughey-O'Kelly condition. The damped system is still linear, but its eigenvectors are complex, indicating that the state of the damped system is not only determined by displacements but also by velocities of discrete nodes.

When the imaginary part of the eigenvector Ψ_r is negligible, $\omega_r \approx \omega_{0r}$, the damped system is approximately equivalent to a combination of n SDOF systems. Furthermore, VIV can be simplified as a single-modal resonance vibration. The resonance response X can be approximated as:

$$X \approx \frac{\Psi_r \Psi_r^T F}{2\zeta_r M_r \omega_r^2} \quad (7)$$

where, $M_r = |\Psi_r^T M \Psi_r|$ is the modal mass, and usually set to 1 to normalize the modal shape vector Ψ_r and F is the excitation force. It indicates that the VIV response is inversely proportional to ζ_r . The generally considered optimization objective of the smallest resonance amplitude can be simplified to the calculation of the maximum complex modal damping ratio ζ_r .

The r -th modal damping ratio ζ_r of a structure with discrete viscous dampers can be approximated as (Main and Krenk, 2005):

$$\zeta_r \approx \frac{\Delta\omega_{\infty r} \eta_r}{\omega_{0r}(1 + \eta_r^2)} \quad (8)$$

where, $\Delta\omega_{\infty r}$ is the frequency increment between undamped system ($c_j = 0$) and constrained system ($c_j = \infty$); η_r is the dimensionless viscous damping parameter defined as:

$$\eta_r = \frac{\Psi_r^T C \Psi_r}{2\Delta\omega_{\infty r}} = \frac{\sum_j c_j \psi_{0r,j}^2}{2\Delta\omega_{\infty r}} \quad (9)$$

where, Ψ_{0r} is the r -th modal shape of the undamped system, and $\psi_{0r,j}$ represents the relative modal displacement between the two positions where the j -th damper connects. When the position of dampers is determined, $\Delta\omega_{\infty r}$ is constant and proportional to $\psi_{0r,j}$ (Main and Krenk, 2005). Fig. 4 shows that ζ_r increases then decreases when η_r increases.

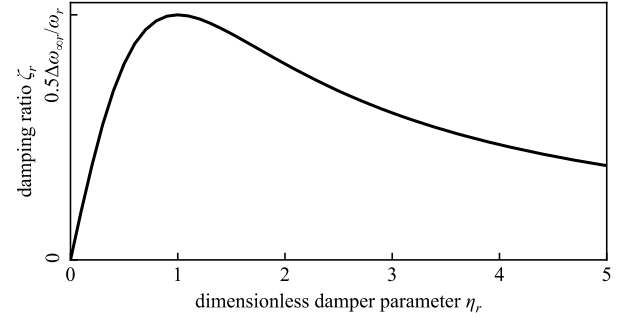


Fig. 4. Unimodal characteristic of ζ_r versus η_r .

When $\eta_r = 1$, ζ_r takes the maximum value. A damper placed in a location with larger $\psi_{0r,j}$ has a larger optimal ζ_r and requires a smaller optimal c_j .

3.2. Parameter contribution and multi-mode optimization

A long-span bridge has dense low-frequency modes, and VIV resonance may occur in multiple modes. For each selected damper location, ζ_r is an unimodal function of the damper parameter c_j . The optimal damper parameter for a single-mode $c_{j,opt}$ can be approximately obtained as:

$$c_{j,opt} = \frac{2\Delta\omega_{\infty r}}{\psi_{0r,j}^2} \quad (10)$$

However, the optimal damper parameter varies from mode to mode. In practice, the most crucial factor is the minimum ζ_r of the considered modes. The optimal objective is to maximize the minimum ζ_r , that is:

$$\text{maximize}\{\min\{\zeta_1, \zeta_2, \dots, \zeta_n\}\} \quad (11)$$

Fig. 5 shows that the minimum function maintains the unimodal characteristics when the derivatives of all the functions are strictly monotonic. In addition, the minimum function is a piecewise function, and the maximum value is located at the functions' intersection or a maximum point of one of the functions, which makes it difficult to obtain the analytical solution.

Ternary search is an iterative algorithm for finding the peak value of such an unimodal function. The algorithm starts with a limited open interval (l, r) , takes two points (a, b) to divide the interval into three parts, and after comparing the function values at a and b , reduces the interval to the two sub-intervals around the maximum, then moves to the next iteration. An optimized version of the ternary search algorithm proposed by Kiefer (1953) is utilized as shown in the dashed box in Fig. 6. The golden ratio, $(\sqrt{5}-1)/2$, is introduced to divide the interval so that the function value can be reused in the next iteration. The algorithm continually narrows the interval proportionally during the iteration and

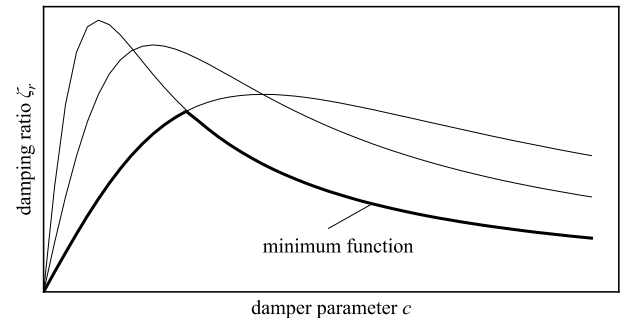


Fig. 5. Unimodal characteristic of the minimum function.

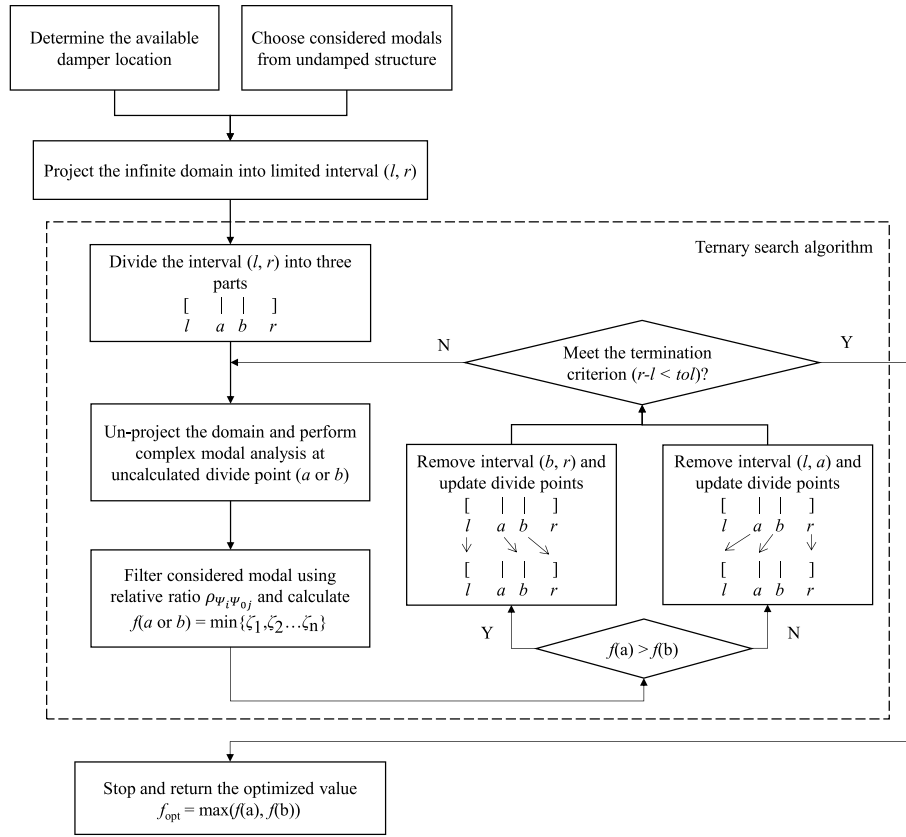


Fig. 6. Flowchart of the multi-mode optimization process for a single parameter.

has been proved to provide the best performance of the ternary search algorithm.

In this study, the argument of the unimodal function is the damper parameter c_j . The initial interval is determined by the available range of c_j . If the available range is the infinite interval $(0, +\infty)$, a substituted parameter can be used to project it to a limited interval, for instance:

$$c_j = c_0 \cdot \tan x \quad (12)$$

where, x is the substituted parameter defined in range $(0, \pi/2)$. The scale parameter c_0 can be set to the single-mode optimal damper parameter by calculating Equation 10.

To programmatically implement the iterative process, the program needs to be able to automatically extract the desired modes. In this paper, we assume that the difference in the modal shapes between the damped and undamped system is negligible, and the relative ratio can be used to quantify the similarity between modes:

$$\rho_{\Psi_i \Psi_{0j}} = \frac{|\Psi_i^T \Psi_{0j}|}{|\Psi_{0j}^T \Psi_{0j}|} \quad (13)$$

where, Ψ_{0j} denotes the j -th modal shape of the undamped system. A damped mode with the largest $\rho_{\Psi_i \Psi_{0j}}$ is considered to correspond to the j -th undamped mode.

Fig. 6 summarizes the multi-mode optimization process for a single damper parameter. The available locations should be first determined based on the conception described in Section 2. Then, the considered modes can be determined based on the modal analysis of the undamped structure. Before performing the iteration, the infinite domain of the damper parameter should be projected into a limited interval using a substituted parameter. For each iteration, complex modal analysis is performed at one of the uncalculated divide points. Then, the considered modes are filtered, the minimum damping ratio is calculated, and the

iteration interval is updated based on the ternary search algorithm. When the interval is smaller than the termination threshold, the maximum value is returned as the optimal damper parameter. In addition, the unimodal characteristic of the minimum function can be considered to simplify the optimization algorithm for multiple damper parameters tuning, and the mutual influence between dampers should be studied in future investigations.

4. Case study

4.1. Structural properties of the Yingwuzhou Yangtze River Bridge

The Yingwuzhou Yangtze River Bridge is located on the Yangtze River in Wuhan, Hubei, China. It has three towers and 4 spans in an arrangement of 200 + 850 + 850 + 200 m, as shown in Fig. 7. The bridge girder is simply supported on the lower beam of the tower. Fig. 8 illustrates the girder section is 3 m high and 38 m wide, consisting of a 20 cm thick concrete deck and steel longitudinal beams. An I-shaped steel diaphragm is installed to connect the longitudinal beams every 3 m along the bridge.

The Yingwuzhou Yangtze River Bridge opened on December 28, 2014. Two perceptible vertical vibrations occurred on April 26, 2020, causing discomfort to the drivers and arousing public attention. During the vibration, the monitoring data showed that the average wind speed was between 7 and 7.6 m/s with small fluctuation ($I_u = 5.8\text{--}7.0\%$), the measured maximum acceleration amplitude was 0.75 m/s^2 located at 3/8 of the main span on the Hanyang side, and the main frequency of the monitored vibration was 0.2441 Hz (Zhu et al., 2021). Since VIV of a bridge always has a quasi-sinusoidal displacement and acceleration, the corresponding maximum amplitude of the displacement can be obtained, $A = 0.75 / (2\pi \times 0.2441)^2 = 0.32 \text{ m}$. To further investigate the vibration characteristic, the first 23 vertical modes of the undamped structure were calculated by the commercial finite element software

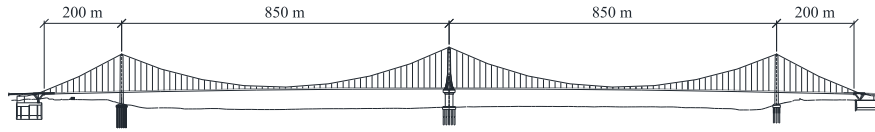


Fig. 7. The elevation view of the Yingwuzhou Yangtze River Bridge.

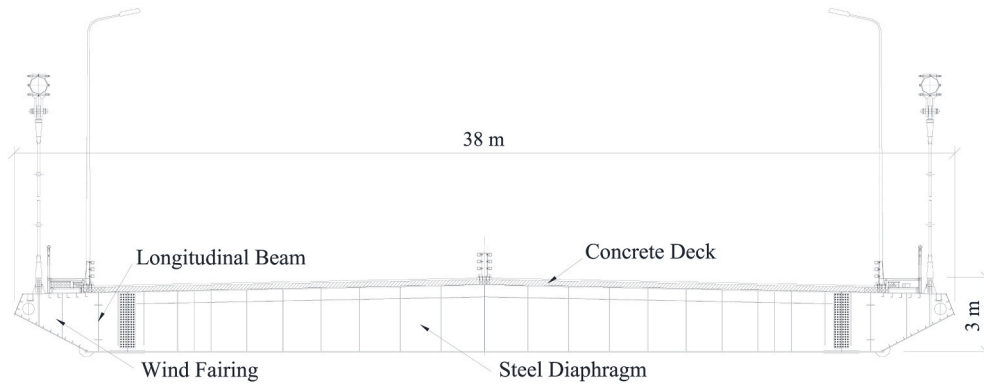


Fig. 8. The section of the bridge girder.

ANSYS and listed in Table 1. The 7th modal frequency ($f_{v7} = 0.2421$ Hz) is close to the vibration frequency, and the monitored vibration shape was close to the 7th vertical modal shape of the structure, in which the amplitude was more significant in 1/8, 3/8, 5/8, and 7/8 span of the two middle spans as shown in Table 1. It was confirmed that the vibration was vortex-induced.

The Strouhal number (St number) is a non-dimensional parameter that indicates the frequency-related VIV starting wind speed. The number corresponding to the monitored VIV event is $St = f_{v7} \times H/U = 0.2441 \times 3/7 = 0.105$. The predicted VIV starting wind speed of each mode was listed in Table 1, and the first 23 modes covered all the modes that possibly suffered VIV events when the wind speed was lower than 25 m/s. The equivalent mass ($m_{eq} = \phi_g^T \phi_g L / M_T$) calculated for 2-DOF test prediction was also addressed in Tables 1 and it was similar in different modes except for some modes coupling the side span vibration. It should be noted that the frequency of the 8th mode was extremely close to that of the 7th mode, so it could be probably excited at a similar wind condition. It seems that more than one mode may be excited by the vortex-induced force at a certain wind velocity since their frequencies are so close. However, due to the phenomenon of lock-in originating from the nonlinearity of VIV, generally only one mode will be excited as observed from field measurement. At present, the VIV lock-in regimes of long-span suspension bridges with closely spaced vertical modes are still unclarified. It may be influenced by many factors, including the time history of wind, the distribution of wind speeds and wind attack along the bridge, the modal characteristics of adjacent modes, the vortex shedding mechanism, and so on. It is still difficult to accurately predict which mode of the bridge the vortex-induced vibration will be lock-in to under a certain wind velocity. To reduce the occurrence of vortex-induced vibration of a long-span suspension bridge, multiple modes should be taken into consideration in VIV control.

4.2. Damper design and parameter optimization

After comprehensive consideration of the structural characteristics and the construction difficulty, 4 dampers were decided to be placed on the two sides of the middle tower. Fig. 9(a) shows the schematic of the damper arrangement. A total of five locations were considered, and the distance between the dampers and the center of the middle tower (D)

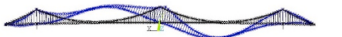
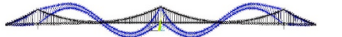
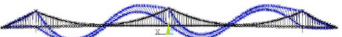
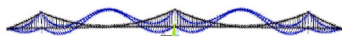
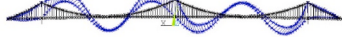
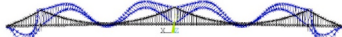
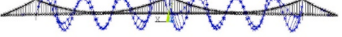
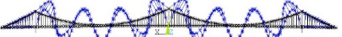
was 5, 8, 11, 17.5, and 20 m, respectively. The location where $D = 20$ m corresponds to the position of the suspenders closest to the middle tower. Due to the symmetrical arrangement, the damper parameters were consistent ($c_j = c$). The contribution of the dampers was modeled by the longitudinal spring-damper elements “Combin14”, which forms the location-related term in the damping matrix. One end of the damper element was attached to the bridge girder, and the other end to the rigid beam extended out of the tower, as shown in Fig. 9(b).

The single-mode optimization was conducted first to examine the influence of the damper parameter c and damper location D on each vertical mode. The iterative optimization algorithm introduced in Section 3 was used to find the optimal parameter. Fig. 10(a) shows the change in the optimal damper parameter c_{opt} with the increase of the vertical modal order, and the change of the corresponding optimal additional modal damping ratio ζ_{opt} were illustrated in Fig. 10(b). It was inspiring that c_{opt} of high-order modal are close, indicating that a set of optimal dampers can simultaneously control multi-mode vibration. Moreover, c_{opt} decreased and ζ_{opt} increased with the increase of the modal order, except for the 4, 6, and 9th modes. The inefficiency of the 4, 6, and 9th modes is because the side spans participated in the modal shape. Under the considered cases, the modal amplitude at the damper location increased with D . When D increased from 5 m to 20 m, c_{opt} became over ten times smaller and ζ_{opt} increased by a factor of 1.5 times.

In the multi-mode optimization, the target modes were chosen based on the following considerations. For modes with low frequency, the lock-in wind speed of VIV is low and narrow. Since high turbulence intensity is unavoidable for low wind speed, VIV rarely occurs in low frequency modes. From the field monitoring of VIVs of the Xihoumen suspension bridge, VIV is more likely to occur at a wind velocity above 5 m/s (Li et al., 2017). On the other hand, for modes with high frequency, the lock-in wind speed of VIV is high, and their occurrence probability is considerably low. Moreover, the operational wind velocity for a long-span bridge is usually below 25 m/s, and it is suggested by the Wind-resistant Design Specification for Highway Bridge of China that only large-amplitude VIV happens below the wind speed of 25 m/s needs to be controlled. Therefore, it is reasonable to choose the target modes corresponding to the VIV wind speed within the range of 5–25 m/s.

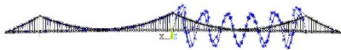
For the case of Yingwuzhou Yangtze River Bridge, it can be seen from

Table 1
The first 23rd vertical modes of the undamped structure.

Mode	Modal shape	Frequency (Hz)	Equivalent mass (t/m)	VIV starting wind speed (m/s)
v1		0.1008	40.58	2.9
v2		0.1277	41.07	3.7
v3		0.1280	41.17	3.7
v4		0.1607	41.49	4.6
v5		0.1827	40.85	5.3
v6		0.2105	52.97	6.1
v7		0.2421	40.94	7.0
v8		0.2421	40.85	7.0
v9		0.2888	107.81	8.4
v10		0.3086	40.83	8.9
v11		0.3162	46.20	9.1
v12		0.3779	40.94	10.9
v13		0.3780	40.94	10.9
v14		0.4526	41.06	13.1
v15		0.4532	41.08	13.1
v16		0.5325	41.30	15.4
v17		0.5325	41.29	15.4
v18		0.6191	41.57	17.9
v19		0.6192	41.57	17.9
v20		0.7126	41.98	20.6
v21		0.7126	41.95	20.6
v22		0.8140	42.42	23.5

(continued on next page)

Table 1 (continued)

Mode	Modal shape	Frequency (Hz)	Equivalent mass (t/m)	VIV starting wind speed (m/s)
v23		0.8140	42.42	23.5

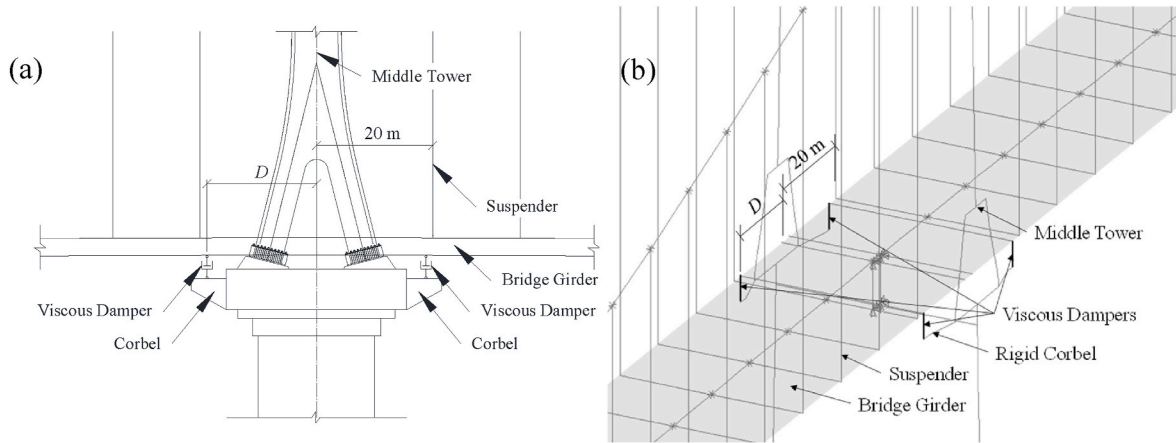
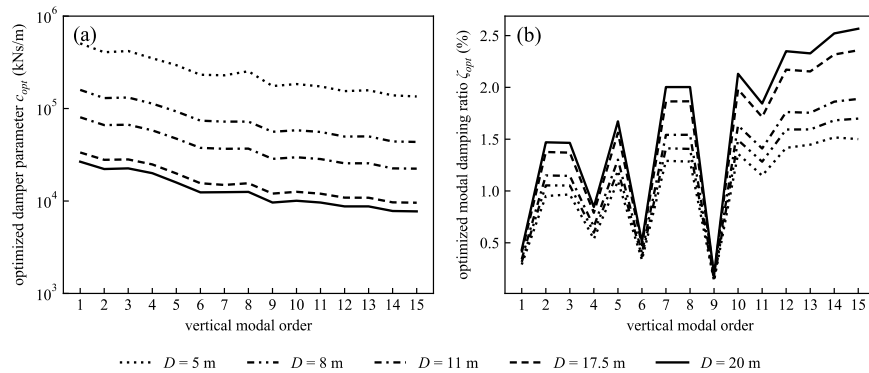
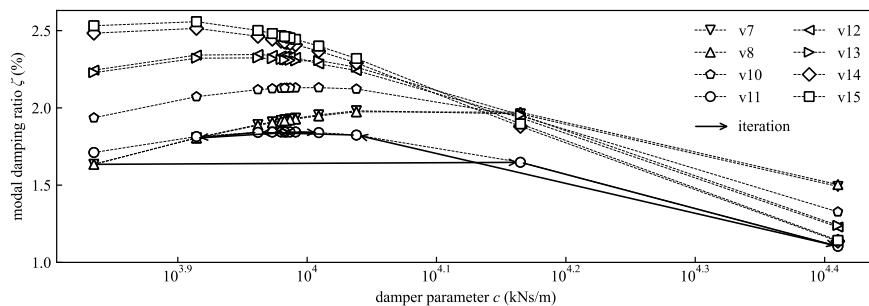
**Fig. 9.** The (a) Schematic and (b) finite element model of the damper arrangement.**Fig. 10.** Optimal (a) damper parameters and (b) modal damping ratio for each vertical mode and damper location.

Table 1 that the vertical bending mode 5 to mode 23 corresponds to the wind speed range of 5–25 m/s. In the design of the damper system, special consideration was given to the 7th mode that experienced VIV during the field monitoring. It was set to be the lower bound in multi-mode optimization to ensure that its additional damping ratio is large enough. For higher modes, more attention was paid to those corresponding to VIV wind speed below 15 m/s because the monitoring data

of wind speed in recent years showed that the wind speed on the bridge deck is mainly below 15 m/s. Moreover, the optimal damper parameters and additional damping ratios for modes higher than the 15th are close based on the parameter analysis. In addition, the modal displacement at the damper position for the 9th vertical mode is so small that dampers showed little efficiency in controlling it, so this mode was not considered in the optimization. Based on the above consideration, multi-mode

**Fig. 11.** Iteration of the multi-mode optimization.

optimization was performed for the 7, 8, and 10-15th vertical modes in the present study.

The dampers were placed under the closest suspenders ($D = 20$ m). The algorithm took thirteen iterations to make the relative accuracy reach 1%, as shown in Figs. 11 and 12. The arrow line illustrates that the iteration started from the left point and converged at the optimal point. In this study, c_{opt} was equitant to the optimal parameter for the 11th mode. Moreover, the additional damping for each considered mode had little variance near the optimal point, suggesting that the optimal damper parameter was stable and robust.

4.3. Wind tunnel test on the effect of damping ratio

To examine the contribution of the damping ratio to the VIV characteristics, a wind tunnel test was conducted in the HD-2 wind tunnel at Hunan University. The structure was simplified to a 2-DOF rigid suspension model based on the similarity of the first vertical and torsional modes. The rigid model was made up of a steel skeleton and wooden cover. A total of eight springs hung the model on the aluminum frame, as shown in Fig. 13. The structure parameters of the model and real bridge are listed in Table 2.

In the present study, the influence of the damping ratio was examined under a wind attack angle of $+3^\circ$, which was the worst condition in the previous experiment. Three laser displacement sensors were adopted to measure the vibration displacements. The incoming wind speed was measured by an ultrasonic wind speed indicator, which is located 2 m in front of the model to eliminate the influence on the wind field. The effect of damper location is ignored because the damper was located near towers in the present study with no change in the sectional characteristic of the bridge girder. The structure damping ratio was adjusted by two plate-typed eddy current dampers, whose damping force can be precisely adjusted by tuning the air gaps (Hua et al., 2022), and the amplitude-related damping characteristic of each damping state is shown in Fig. 14. A total of 6 damping states were examined in the present study. The damping ratio was observed to increase slightly with the vibration amplitude. It should be noted that the damping characteristic was different between the sectional model and the real bridge. The value corresponding to the peak VIV amplitude was used as the nominal damping ratio in the present study which overestimated the structural damping, and the resulting response was expected located under the safety side.

Fig. 15(a) illustrates the vertical response amplitude of the wind speed under different damping ratios. The effect of the modal shape was considered by multiplying a correction factor of 1.15 based on the previous studies (Zhang et al., 2014). A dashed line was appended to indicate the maximum allowed amplitude of the 7th mode by the Chinese specification (Ministry of Communications of the People's Republic of China, 2018). Two VIV lock-in interval was observed in a lower

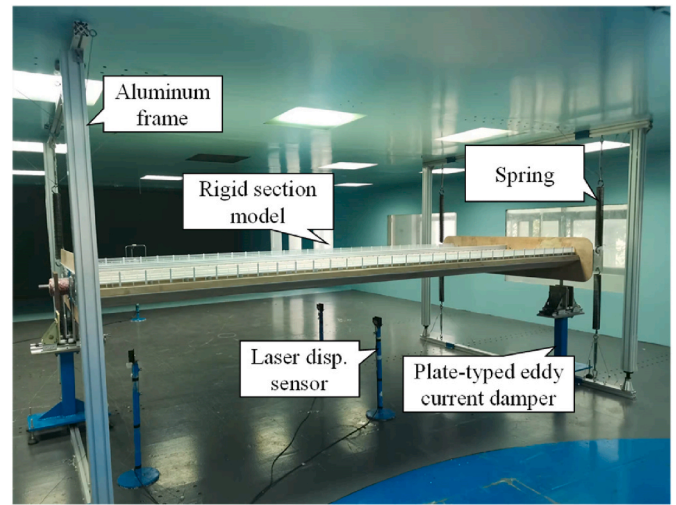


Fig. 13. Photo of the wind tunnel experiment.

Table 2

Parameter comparison between the model and actual bridge.

Parameter	Actual	Scale ratio	Model
Rigid model length L (m)	/	/	3.78
Girder width B (m)	38	1/25	1.52
Girder height H (m)	3	1/25	0.12
Equivalent mass m_{eq} (kg/m)	40581.3	1/25 ²	64.93
Equivalent inertia momentum I_{eq} (kg·m ² /m)	5759780	1/25 ⁴	14.75
Frequency of the 1st vertical mode f_{v1} (Hz)	0.101	1/23.9	2.411

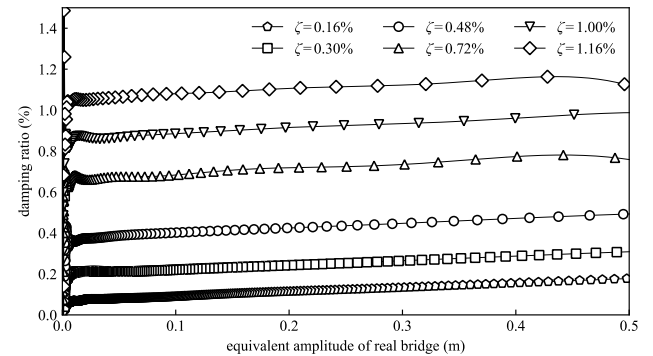


Fig. 14. The amplitude-related damping characteristics of the sectional model.

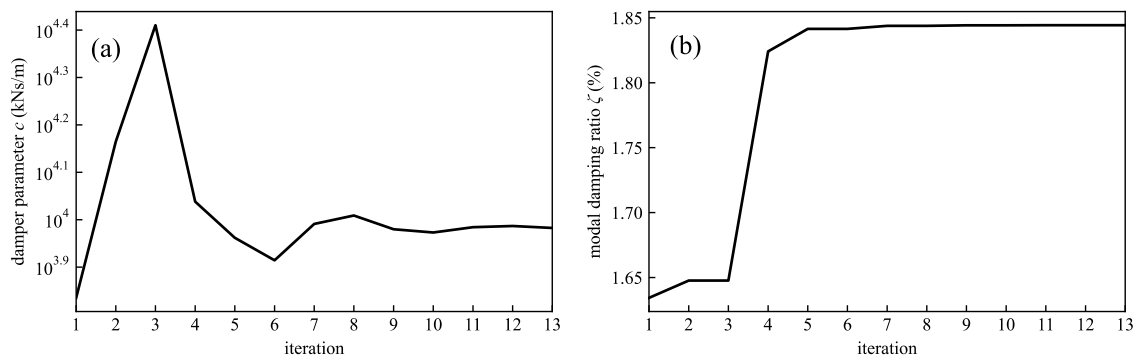


Fig. 12. Multi-mode optimization convergence curve of (a) the damping parameter and (b) the minimum modal damping ratio.

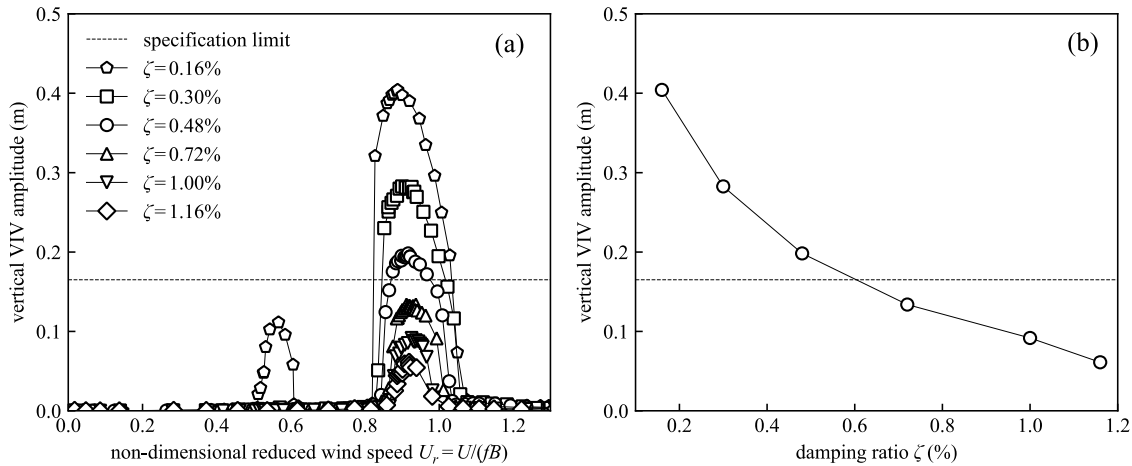


Fig. 15. The vertical VIV amplitude versus (a) wind speed and (b) damping ratio.

damping ratio, representing two St numbers, $St_1 = 0.152$ and $St_2 = 0.098$. The small interval with lower wind speed quickly disappeared when the damping ratio increased. The other interval with higher wind speed had a similar St number with the monitored VIV event ($St = f_{V7} \times H/U = 0.2441 \times 3/7 = 0.105$), and the VIV amplitude had an approximately inversely proportional relation to the damping ratio, as shown in Fig. 15(b). To satisfy the specification requirement, the damping ratio must be larger than 0.6% in the real bridge.

4.4. Field installation and measurement

Based on the aboved numerical examination, the dampers' designed parameter c was set to $1 \times 10^4 \text{ kNs/m}$ and the dampers were placed under the first suspenders (Hua et al., 2023). The additional damping ratio of each vertical mode is listed in Table 3. All the mode with VIV starting wind speed lower than 25 m/s was validated. Similar to that in the cable's vibration control, the higher mode is expected to have a better control effect for its higher additional damping ratio.

It should be addressed that the modal characteristics are slightly different between structures with or without dampers. For the 7th vertical mode in which VIV occurred, the modal frequency increases from 0.2421 Hz to 0.2456 Hz, and the difference in the modal shape is illustrated in Fig. 16. The modal shape of the damped structure consists of real and imaginary parts. The real part is similar to the undamped modal shape and turning points exist at the damper locations ($D = \pm 20 \text{ m}$). Different from the structure without dampers, there is no zero-magnitude stationary point in the modal shape of the damped structure due to the contribution of the imaginary part of the complex modal shape. The difference of modal frequency and modal shape was ignored in the present study and the modal damping ratio is considered as the main factor for field measurement.

Table 3
Numerical predicted additional damping ratio of each vertical mode.

Mode	Numerical predicted additional damping ratio	Mode	Numerical predicted additional damping ratio	Mode	Numerical predicted additional damping ratio
v1	0.282%	v9	0.191%	v17	2.490%
v2	1.079%	v10	2.131%	v18	2.497%
v3	1.064%	v11	1.842%	v19	2.533%
v4	0.667%	v12	2.295%	v20	2.512%
v5	1.487%	v13	2.317%	v21	2.544%
v6	0.496%	v14	2.389%	v22	2.493%
v7	1.944%	v15	2.422%	v23	2.537%
v8	1.939%	v16	2.469%		

In this project, a type of torsional eddy current damper (TECD) is adopted. Fig. 17 shows the construction and figure of the damper. The connecting rods translate the vertical vibration and release the constraint of the lateral and torsional displacement. The translated vertical vibration is then amplified by the gear system and then dissipated by the torsional damper. Key parameters of the TECD are listed in Table 4. The TECD only weighs about 600 kg, which is a thousand times less than the additional mass required for TMD (calculated by 1% of the modal mass of the main spans, that is, $1\% \times 40 \text{ t} \times 1700 \text{ m} = 680 \text{ t}$). Compared to the traditional oil damper, TECD can provide a large damping coefficient under low vibration speed. Its metal structure requires low maintenance costs and has no oil leakage problems.

The structural dynamic characteristics were measured under the condition of closed traffic for five nights from July 24 to July 28, 2021. The environment wind speed was relatively small, no VIV event happened, and the contribution of the aerodynamic damping was ignored. The measurement compared the structural damping ratio under the conditions with or without dampers. To enlarge the response amplitude and improve the measurement accuracy, the continuous vehicle bumping excitation method was used for excitation (Hua et al., 2017). Since the frequency of the symmetry and anti-symmetry modes were close, the monitors were installed on the half side of the bridge. A total of 36 wireless acceleration sensors were installed to monitor the vibration characteristics. Fig. 18 shows a velocity time history obtained in an excitation experiment without or with dampers. The background vibration was suppressed after the damper was installed. The stochastic subspace identification (SSI) technique was adopted to extract the modal frequencies and the modal damping ratios. The SSI method has a complete theory and powerful algorithm and is one of the most effective modal parameter recognition methods based on environmental vibration or free vibration. The extracted results were averaged over repeated experiments of different excitation modes and listed in Table 5. The extracted modes were labeled by the closest anti-symmetry modes in the finite element model. The extracted damping ratio of the structure without dampers was between 0.25% and 0.38%, and the theoretical VIV amplitudes of all the modes exceeded the specification limits. For the bridge with dampers, the modal damping ratio is between 0.64% and 1.11%. For the 7th vertical model in which VIV occurred, the additional damping ratio increased from 0.34% to 1.00% and the predicted amplitude of VIV met the requirements of the specification. In addition, the VIV locking interval was reduced to half of the original one based on the result of the wind tunnel test, and the probability of VIV occurrence should be greatly reduced. Meanwhile, the structure health monitoring system has been updated after the VIV event, and three years have passed since the dampers were installed, and no VIV has occurred again.

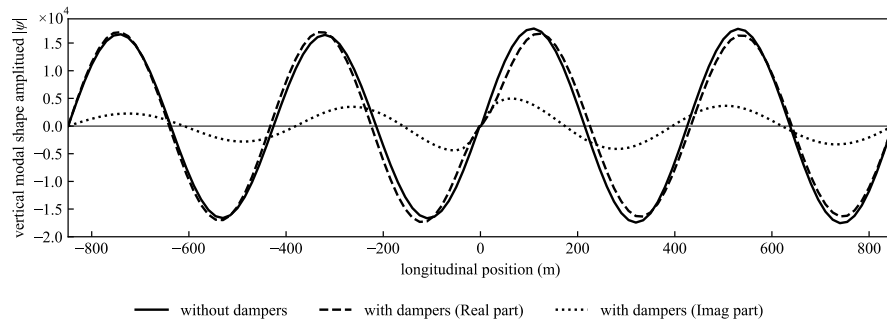


Fig. 16. The modal shape amplitude difference of structures with or without dampers.

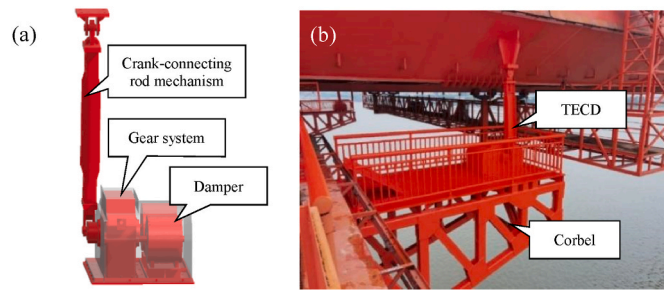


Fig. 17. The (b) construction and (c) photo of the torsional eddy current damper.

Table 4
Key parameters of the TECD.

Size (mm)	Weight (kg)	Working stroke (mm)	Maximum damping force (kN)	Damping Coefficient (kNs/m)
1600 × 1200	600	±80	500	10000

on the bridge. Though a further track should be conducted to validate the control performance of the dampers, the solution is considered a feasible approach in VIV controlling for long-span suspension bridges. However, the measured damping ratio was much smaller than the numerical results, especially for the high-order modes. Similar results have been observed and reported by Zoltowski et al. (2022). One possible reason was that the structural gaps and frictions of the dampers and the bridge supports made the structural system inconsistent with the numerical model. In addition, the vibration amplitude excited by the jump load was too small to fully unleash the performance of the dampers. For example, the static friction may lock the support's rotation and the bridge pin support will change to a fixed support. The modal shape will hence change and reduce the modal displacement at the dampers'

location. It suggests that the local characteristic between dampers and bridge supports plays a significant role in the dampers' performance. A further examination is required to explore the inconsistency.

5. Conclusions

In the present study, a discrete linear viscous damper solution is proposed to suppress the multi-mode VIV response of the long-span suspension bridges. The construction concept and the optimization process of the damping system are fully addressed, and an application on the Yingwuzhou Yangtze River Suspension Bridge is presented in detail as an illustration. The main conclusions are as follows:

1. Viscous dampers near the support of girder are a viable method to increase the modal damping of long-span suspension bridges and suppress vertical VIV, provided there is enough distance from the support to the point of attachment of the damper. The maximum possible modal damping increases with this distance.
2. For a specified location, the maximum possible damping can be obtained by employing an optimal damper size, which varies with the mode number. Fortunately, the difference in optimal damper size gradually decreases as the increasing mode number, which indicates that the viscous damper can robustly control multi-mode VIV

Table 5
The monitored modal frequency and damping ratio without or with dampers.

Mode	Frequency (Hz)	Damping ratio without dampers (%)	Damping ratio with dampers (%)	Additional damping ratio (%)	Numerical predicted additional damping ratio (%)
v1	0.105	0.37	0.74	0.37	0.282%
v3	0.135	0.29	0.65	0.36	1.064%
v5	0.187	0.25	0.64	0.39	1.487%
v7	0.245	0.34	1.00	0.66	1.944%
v11	0.318	0.38	1.11	0.73	1.842%
v13	0.390	0.31	0.72	0.41	2.317%

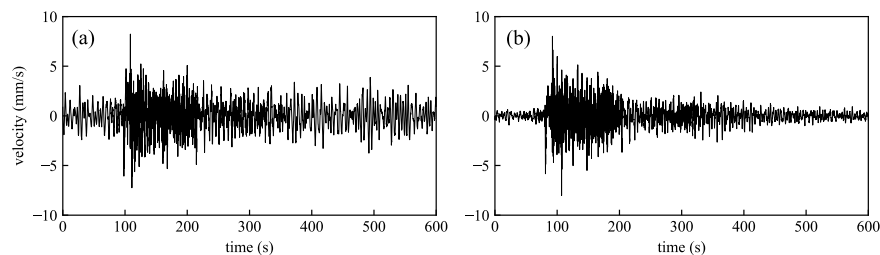


Fig. 18. Velocity time history during excitation (a) without or (b) with dampers.

simultaneously. However, a damper size optimized for the first several modes may result in ineffectiveness for controlling VIV in higher-order modes.

3. A design procedure for viscous dampers for damping of multiple modes, corresponding to vertical VIVs that occur in the wind speed range of 5~25 m/s, has been presented. In this method, the optimal damper size is obtained by maximizing the minimum modal damping ratio of all target modes. Such an optimization is able to provide enough damping capacity for all assigned modes of the suspension bridge.
4. The damping system has been applied successfully to the Yingwuzhou Yangtze River Suspension Bridge. The measured damping ratios for some critical modes are increased by three times after installation of the dampers. As for the mode that experienced VIV, its modal damping ratio has been raised high enough to suppress the VIV response according to the sectional model wind tunnel test results and the requirement of Wind-resistant Design Specification for Highway Bridges of China. Though a long-term monitoring should be conducted to validate damper's performance, three years have passed since the dampers were installed, and no VIV event has occurred again to the bridge.
5. It should be noted that there is still a gap between the additional damping ratios obtained from the field measurement and those from the theoretical analysis. Many factors may influence this gap, including the support constraints in real bridge, a mechanical gap in the connection of the damper, and so on. Meanwhile, the sectional modal test cannot consider the contribution of the modal shape and 3D effect of the real bridge. Aeroelastic model testing should be examined in the future to validate and study the different performances between the model and the real bridge and the effect on the different wind conditions.

CRedit authorship contribution statement

Yiwen Cao: Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Zhiwen Huang:** Methodology, Formal analysis, Validation, Visualization, Writing – review & editing. **Hongyi Zhang:** Validation, Investigation, Data curation, Writing – review & editing. **Xugang Hua:** Conceptualization, Methodology, Formal analysis, Validation, Writing – review & editing. **Zhengqing Chen:** Conceptualization, Methodology, Validation, Writing – review & editing, Supervision. **Tianbao Wan:** Validation, Data curation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Bai, H., Ji, N., Xu, G., Li, J., 2020. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: sealed traffic barrier. *J. Wind Eng. Ind. Aerod.* 206, 104302 <https://doi.org/10.1016/j.jweia.2020.104302>.
- Battista, R.C., Pfeil, M.S., 2000. Reduction of vortex-induced oscillations of Rio-Niterói bridge by dynamic control devices. *J. Wind Eng. Ind. Aerod.* 84, 273–288. [https://doi.org/10.1016/S0167-6105\(99\)00108-7](https://doi.org/10.1016/S0167-6105(99)00108-7).
- Caughey, T.K., O'Kelly, M.E.J., 1965. Classical normal modes in damped linear dynamic systems. *J. Appl. Mech.* 32, 583–588. <https://doi.org/10.1115/1.3627262>.
- Chen, L., Sun, L., Xu, Yuyuan, Di, F., Xu, Yingmei, Wang, L., 2020. A comparative study of multi-mode cable vibration control using viscous and viscoelastic dampers through field tests on the Sutong Bridge. *Eng. Struct.* 224, 111226 <https://doi.org/10.1016/j.engstruct.2020.111226>.
- Dai, J., Xu, Z.-D., Gai, P.-P., Hu, Z.-W., 2021. Optimal design of tuned mass damper inerter with a Maxwell element for mitigating the vortex-induced vibration in bridges. *Mech. Syst. Signal Process.* 148, 107180 <https://doi.org/10.1016/j.ymssp.2020.107180>.
- Diana, G., Resta, F., Belloli, M., Rocchi, D., 2006. On the vortex shedding forcing on suspension bridge deck. *J. Wind Eng. Ind. Aerod.* 94, 341–363. <https://doi.org/10.1016/j.jweia.2006.01.017>.
- Ehsan, F., Scanlan, R.H., 1990. Vortex-induced vibrations of flexible bridges. *J. Eng. Mech.* 116, 1392–1411. [https://doi.org/10.1061/\(ASCE\)0733-9399\(1990\)116:6\(1392\)](https://doi.org/10.1061/(ASCE)0733-9399(1990)116:6(1392)).
- Frandsen, J.B., 2001. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. *J. Wind Eng. Ind. Aerod.* 89, 95–129. [https://doi.org/10.1016/S0167-6105\(00\)00059-3](https://doi.org/10.1016/S0167-6105(00)00059-3).
- Fujino, Y., Yoshida, Y., 2002. Wind-induced vibration and control of Trans-Tokyo Bay crossing bridge. *J. Struct. Eng.* 128, 1012–1025. [https://doi.org/10.1061/\(ASCE\)0733-9445\(2002\)128:8\(1012\)](https://doi.org/10.1061/(ASCE)0733-9445(2002)128:8(1012)).
- Giaralis, A., Petrin, F., 2017. Wind-induced vibration mitigation in tall buildings using the tuned mass-damper-inerter. *J. Struct. Eng.* 143, 04017127 [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0001863](https://doi.org/10.1061/(ASCE)ST.1943-541X.0001863).
- Gu, M., Chen, S.R., Chang, C.C., 2002. Control of wind-induced vibrations of long-span bridges by semi-active lever-type TMD. *J. Wind Eng. Ind. Aerod.* 90, 111–126. [https://doi.org/10.1016/S0167-6105\(01\)00165-9](https://doi.org/10.1016/S0167-6105(01)00165-9).
- Hua, X., Ma, W., Huang, Z., Chen, Z., Wan, T., Sun, Y., Liu, S., 2022. Effect of Sc number on vertical vortex-induced vibration of bridges based on precise damping adjustment. *J. Hunan Univ., Nat. Sci.* 49, 17–25. <https://doi.org/10.16339/j.cnki.hdxzkb.2022045>.
- Hua, X., Zhang, J., Wan, T., Sun, Y., Chen, Z., 2023. A tower-girder damping system to control multi-mode vertical vortex-induced vibrations of suspension bridge. *Bridge Constr.* 53, 16–24. <https://doi.org/10.20051/j.issn.1003-4722.2023.04.003>.
- Hua, X., Zhou, Y., Yang, K., Wen, Q., Chen, Z., Li, Y., 2017. Damping identification of long-span bridges from continuous vehicle bumping excitations. *J. Railw. Sci. Eng.* 14, 1664–1673. <https://doi.org/10.19713/j.cnki.43-1423/u.2017.08.013>.
- Huang, Z., Hua, X., Chen, Z., Niu, H., 2019. Optimal design of TVMD with linear and nonlinear viscous damping for SDOF systems subjected to harmonic excitation. *Struct. Control Health Monit.* 26 <https://doi.org/10.1002/stc.2413>.
- Jones, N.P., Scanlan, R.H., Sarkar, P.P., Singha, L., 1995. The effect of section model details on aeroelastic parameters. *J. Wind Eng. Ind. Aerodyn., Third Asian-Pacific Sympos. Wind Eng.* 54–55, 45–53. [https://doi.org/10.1016/0167-6105\(94\)00028-C](https://doi.org/10.1016/0167-6105(94)00028-C).
- Kiefer, J., 1953. Sequential minimax search for a maximum. *Proc. Am. Math. Soc.* 4, 502–506. <https://doi.org/10.1090/S0002-9939-1953-0055639-3>.
- Krenk, S., 2004. Complex modes and frequencies in damped structural vibrations. *J. Sound Vib.* 270, 981–996. [https://doi.org/10.1016/S0022-460X\(03\)00768-5](https://doi.org/10.1016/S0022-460X(03)00768-5).
- Kubo, Y., Kimura, K., Sadashima, K., Okamoto, Y., Yamaguchi, E., Kato, K., 2002. Aerodynamic performance of improved shallow p shape bridge deck. *J. Wind Eng. Ind. Aerod.* [https://doi.org/10.1016/S0167-6105\(02\)00328-8](https://doi.org/10.1016/S0167-6105(02)00328-8).
- Laima, S., Li, H., Chen, W., Ou, J., 2018. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. *J. Fluid Struct.* 77, 115–133. <https://doi.org/10.1016/j.jfluidstructs.2017.12.005>.
- Larose, G.L., D'Auteuil, A., 2006. On the Reynolds number sensitivity of the aerodynamics of bluff bodies with sharp edges. *J. Wind Eng. Ind. Aerodyn., Eighth Ital. Nat. Conf. Wind Eng. IN-VENTO-2004* 94, 365–376. <https://doi.org/10.1016/j.jweia.2006.01.011>.
- Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, T., 2000. Storebælt suspension bridge – vortex shedding excitation and mitigation by guide vanes. *J. Wind Eng. Ind. Aerodyn., Int. Conf. wind Eng.* 88, 283–296. [https://doi.org/10.1016/S0167-6105\(00\)00054-4](https://doi.org/10.1016/S0167-6105(00)00054-4).
- Li, S., Laima, S., Li, H., 2017. Cluster analysis of winds and wind-induced vibrations on a long-span bridge based on long-term field monitoring data. *Eng. Struct.* 138, 245–259. <https://doi.org/10.1016/j.engstruct.2017.02.024>.
- Main, J.A., Krenk, S., 2005. Efficiency and tuning of viscous dampers on discrete systems. *J. Sound Vib.* 286, 97–122. <https://doi.org/10.1016/j.jsv.2004.09.022>.
- Ministry of Communications of the People's Republic of China, 2018. Wind-resistant Design Specification for Highway Bridges. China Communications Press, Beijing, China. JTG/T 3360-01-2018.
- Møller, R.N., Krenk, S., Svendsen, M.N., 2019. Damping system for long-span suspension bridges. *Struct. Control Health Monit.* 26 <https://doi.org/10.1002/stc.2448>.
- Schewe, G., Larsen, A., 1998. Reynolds number effects in the flow around a bluff bridge deck cross section. *J. Wind Eng. Ind. Aerod.* 74–76, 829–838. [https://doi.org/10.1016/S0167-6105\(98\)00075-0](https://doi.org/10.1016/S0167-6105(98)00075-0).

- Simiu, E., Scanlan, R.H., 1996. *Wind Effects on Structures: Fundamentals and Applications to Design*, third ed. John Wiley, New York.
- Sun, Z., Zou, Z., Ying, X., Li, X., 2020. Tuned mass dampers for wind-induced vibration control of Chongqi bridge. *J. Bridge Eng.* 25, 05019014 [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001510](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001510).
- Weber, F., Maślanka, M., 2012. Frequency and damping adaptation of a TMD with controlled MR damper. *Smart Mater. Struct.* 21, 055011 <https://doi.org/10.1088/0964-1726/21/5/055011>.
- Xu, K., Bi, K., Ge, Y., Zhao, L., Han, Q., Du, X., 2020. Performance evaluation of inerter-based dampers for vortex-induced vibration control of long-span bridges: a comparative study. *Struct. Control Health Monit.* 27 <https://doi.org/10.1002/stc.2529>.
- Xu, K., Bi, K., Han, Q., Li, X., Du, X., 2019. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: analytical study. *Eng. Struct.* 182, 101–111. <https://doi.org/10.1016/j.engstruct.2018.12.067>.
- Xu, K., Dai, Q., Bi, K., Fang, G., Ge, Y., 2022. Closed-form design formulas of TMDI for suppressing vortex-induced vibration of bridge structures. *Struct. Control Health Monit.* <https://doi.org/10.1002/stc.3016>.
- Yang, C., Chen, Z., Wang, W., Hua, X., Wang, Y., 2021. Optimal design of two viscous dampers for multi-mode control of a cable covering broad frequency range. *Eng. Struct.* 245, 112830 <https://doi.org/10.1016/j.engstruct.2021.112830>.
- Yang, Y., Zhou, R., Ge, Y., Zhang, L., 2016. Experimental studies on VIV performance and countermeasures for twin-box girder bridges with various slot width ratios. *J. Fluid Struct.* 66, 476–489. <https://doi.org/10.1016/j.jfluidstructs.2016.08.010>.
- Zhang, Z., Ge, Y., Chen, Z., 2014. Vortex-induced oscillations of bridges: theoretical linkages between sectional model tests and full bridge responses. *Wind Struct. Int. J.* 19, 233–247. <https://doi.org/10.12989/was.2014.19.3.233>.
- Zhao, L., Cui, W., Shen, X., Xu, S., Ding, Y., Ge, Y., 2022. A fast on-site measure-analyze-suppress response to control vortex-induced-vibration of a long-span bridge. *Structures* 35, 192–201. <https://doi.org/10.1016/j.istruc.2021.10.017>.
- Zhu, J., Xiong, Z., Xiang, H., Huang, X., Li, Y., 2021. Ride comfort evaluation of stochastic traffic flow crossing long-span suspension bridge experiencing vortex-induced vibration. *J. Wind Eng. Ind. Aerod.* 219, 104794 <https://doi.org/10.1016/j.jweia.2021.104794>.
- Zoltowski, K., Banas, A., Binczyk, M., Kalitowski, P., 2022. Control of the bridge span vibration with high coefficient passive damper. Theoretical consideration and application. *Eng. Struct.* 254, 113781 <https://doi.org/10.1016/j.engstruct.2021.113781>.